

Proof Strategy for Predicate Logic

- The general strategy for doing proofs in predicate logic incorporates the strategy for proofs in sentential logic.
 - If you have a *Universal* sentence, use Universal Elimination to derive one or more instances as needed.
 - If you have an existential formula, assume an instance (for Existential Elimination).
 - Apply rules until you get to the desired conclusion or need to make an assumption.
 - If you need to make an assumption, *look at what you are trying to prove* and use the hints above in conjunction with these below:
 - **UNIVERSAL:** Try to prove an instance of it (and then do Universal Introduction -- beware of conditions!)
 - **EXISTENTIAL:** Either assume its negation (for RAA) or try to prove an instance of it (and then do Existential Introduction)

Practical Application of Logic

Analysis of Legal Argument

It is about arguments

In this class we concern ourselves with arguments, that is with collections of statements, which we term “premises,” that are meant to convince one of the veracity of a specific claim, which we term the “conclusion.” While we have focused on symbolic representations, in order to see similarity between arguments, we should not forget that the symbols do represent arguments that can be expressed in English. It is this fact that makes this material a valuable tool for analysis.

Legal Arguments

For this exercise we will examine a legal argument. Why? Because whether we like it or not we are a people of laws. Laws define us, and dictate much of our day to day activity, e.g. laws on speed limits dictate the acceptable speed at which you drove to school/work today.



Kevin Lomax: Why the law? ... Why the lawyers? Why the law?

John Milton: Because the law, my boy, puts us into everything. It's the ultimate backstage pass.

Lee vs. Weisman

- ***Lee v. Weisman***, 505 U.S. 577 (1992), was a United States Supreme Court decision regarding school prayer. It was the first major school prayer case decided by the Rehnquist Court. It involved prayers led by religious authority figures at public school graduation ceremonies. The Court followed a broad interpretation of the Establishment Clause that had been standard for decades at the nation's highest court.

Translation Scheme

Lx = x is a Law

Cx = x is made by Congress

Rx = x endorses Religion

Px = x Prohibits the free exercise of religion

Gx = x is an involuntary action or event which is
Government Sanctioned

a = the Nathan Bishop Middle School
promotional ceremony of June 1989

The Argument

1	(1) $\forall x((Lx \ \& \ Cx) \rightarrow (\sim Rx \ \& \ \sim Px))$	A
2	(2) $\forall x (Gx \rightarrow (\sim Rx \ \& \ \sim Px))$	A
3	(3) $Ga \ \& \ Ra$	A
3	(4) Ga	3 &E
2	(5) $(Ga \rightarrow (\sim Ra \ \& \ \sim Pa))$	2 $\forall$ E
2,3	(6) $\sim Ra \ \& \ \sim Pa$	4,5 $\rightarrow$ E
2,3	(7) $\sim Ra$	6 &E
3	(8) Ra	3 &E
2	(9) $\sim (Ga \ \& \ Ra)$	6,7 RAA (3)

Some notes on 1 and 2

1 is simply a symbolic representation of the “establishment clause” of the First Amendment of the Constitution, i.e. “Congress shall make no law respecting an establishment of religion, or prohibiting the free exercise thereof.”

2 is an interpretation of 1, the “establishment clause,” which resulted from *Everson v. Board of Education*, and built a strict “wall of separation” between church and state.

What does this demonstrate?

If 3 is a correct statement of affairs, then a contradiction results at lines 7 and 8, **and this means that the event in question is inconsistent with the “establishment clause” of the first amendment of the Constitution as interpreted by the Court in previous decisions.**

Why, because it means the assumption at line 3 is inconsistent with the given premises, which are nothing more than representations of the “establishment clause.”

Reasons for accepting the truth of Ga

- A group of faculty members planned a.
- The Principal approved the **faculty's decision** to invite the Rabbi
- The Principal gave the Rabbi a pamphlet concerning the composition of non-sectarian prayer, thereby taking an **active role** in the composition of the *Invocation* and *Benediction*.
- Graduation ceremonies mark a significant moment in a young person's life, and cannot be considered as an event at which one's attendance is in any sense "voluntary."

Reasons for accepting the truth of Ra

- Invocations and Benedictions are by their very nature religious exercises.
- The claim that the Invocation and Benediction were ***necessary*** to “solemnize” the ceremony implies the entire event can be classified as religious in nature due to this *necessary* religious component.
- By taking an active role in the composition of the Invocation and Benediction, through the distribution of a pamphlet composed by clergy of an exclusively Judeo-Christian (Western) tradition, the administration sent a message of endorsement of that tradition over Non-Western traditions, as well as over abstinence from religion altogether.

Models

Finite Interpretations

- **Finite interpretations** play the same role for predicate logic that truth tables did for sentential logic: they give us a way to talk about truth and falsehood for wffs. A finite interpretation is always a finite interpretation **for some set of sentences**.

Three Basic Notions

- **UNIVERSE**
- **PREDICATE EXTENSIONS**
- **TRUTH VALUE SPECIFICATIONS**

Universe

- **UNIVERSE:** A finite set of objects (also called the **DOMAIN** of the interpretation). It must contain at least one object.

The universe for an interpretation can be absolutely any set of objects whatever, so long as it contains at least one thing and only finitely many things.

Examples:

- The set of all states in the United States.
- Chicago, the planet Uranus, the Atlantic Ocean, and my cat Mungo.
- The set of all living US Presidents and former presidents (which is the same set as the set containing Barack Obama, George W. Bush, Bill Clinton, George H. W. Bush, and Jimmy Carter).
- The set of all positive integers less than 100.
- *To indicate a set, we will use curly braces "{}" and list the members of the set within them.*

Predicate Extensions

- **PREDICATE EXTENSIONS:** For each predicate F found in the sentences, the **extension** of F is the set of those things in the universe to which F applies.

(Strictly speaking, this is the definition of the extension of a **one-place** predicate. We can define the extensions of two-place, three-place, etc., predicates, but it requires a more complicated definition. For the moment we will pass over this.)

Truth Value Specifications

- **TRUTH VALUE SPECIFICATIONS:** The finite interpretation must assign a truth value (either true or false) to each sentence letter in the set of sentences. (If the set of sentences does not contain any sentence letters, then there will not be any truth value assignments.)

How does it work?

- Consider this argument:
- $\forall x(Fx \rightarrow Gx), \exists yFy \vdash \forall xGx$
- It claims:
- Everything that is an F is also a G
- Something is an F
- Therefore everything is a G

Is this a valid argument?

- $\forall x(Fx \rightarrow Gx), \exists yFy \vdash \forall xGx$
- We can show it as invalid by constructing a countermodel.
- We first construct a finite model, and then employ the same technique that we used to construct indirect truth tables in chapter 2 to test for invalidity.

$U = \{a, b\}$

$(Fa \rightarrow Ga) \ \& \ (Fb \rightarrow Gb), \ Fa \vee Fb \vdash Ga \ \& \ Gb$

How did you get that?

$(Fa \rightarrow Ga) \& (Fb \rightarrow Gb), Fa \vee Fb \vdash Ga \& Gb$

By expanding the quantified formulas on a given domain.

$U = \{a, b\}$

$\forall x(Fx \rightarrow Gx),$

$\begin{array}{ccc} a & \& & b \\ \swarrow & & \searrow & \searrow \\ (Fa \rightarrow Ga) & \& & (Fb \rightarrow Gb), \end{array}$

$\exists yFy$

$\begin{array}{ccc} a & \vee & b \\ \downarrow & & \downarrow \\ Fa & \vee & Fb \end{array}$

$\vdash \forall xGx$

$\begin{array}{ccc} a & \& & b \\ \downarrow & & \downarrow \\ Ga & \& & Gb \end{array}$

Is this a valid argument?

$U = \{a, b\}$

$(Fa \rightarrow Ga) \ \& \ (Fb \rightarrow Gb), \ Fa \vee Fb \vdash Ga \ \& \ Gb$

T	T	F	F	T	F	T	F
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T

T

T

T

F

Clearly it is possible for the premises to be true, when the conclusion is false, and we can take these assigned values to craft the countermodel.

The countermodel?

$(Fa \rightarrow Ga) \ \& \ (Fb \rightarrow Gb), \ Fa \vee Fb \vdash Ga \ \& \ Gb$

T T F F T F T F

T

T

T

T

F

$U = \{a, b\}$

$F = \{a\}$

$G = \{a\}$